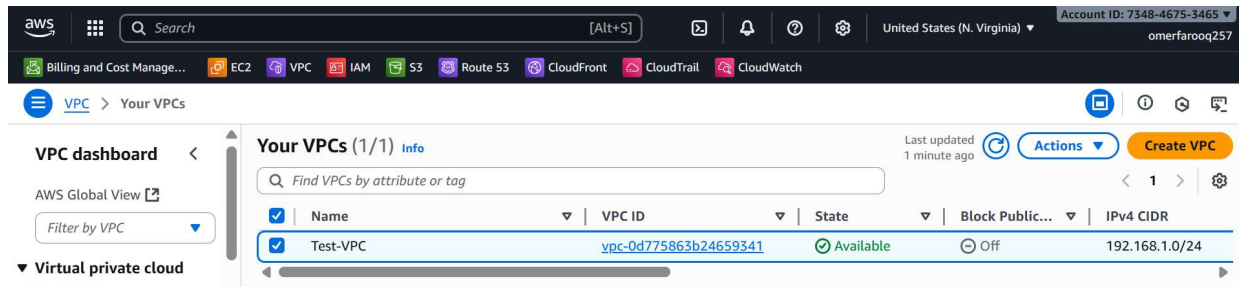


VPC Tasks – Peering, Transit Gateway & Endpoint

1. Create one VPC with one public subnet and one private subnet.

Purpose: To establish a basic multi-tier architecture with public-facing and private resources

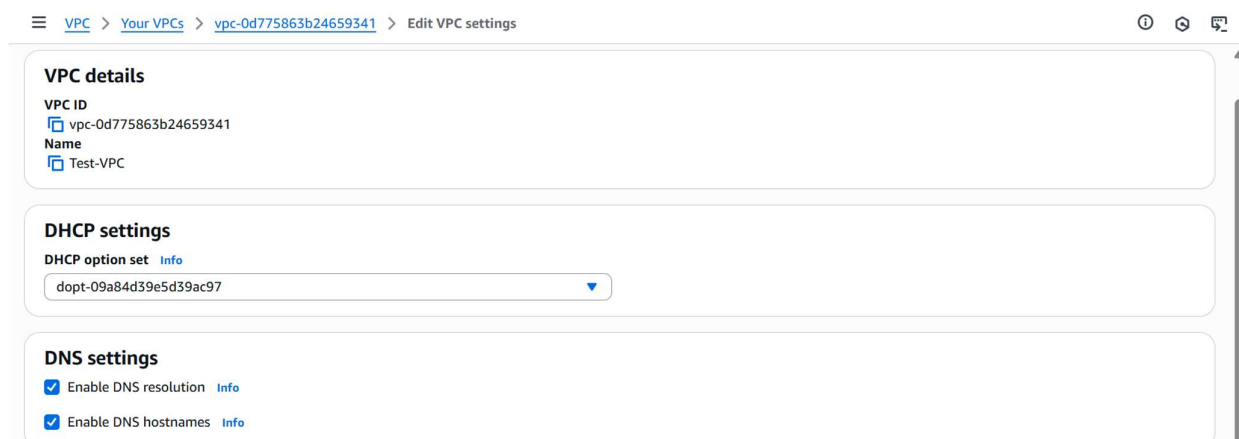
STEPS TO CREATE A VPC WITH ONE PUBLIC AND ONE PRIVATE SUBNET:



Creating a VPC:

- Navigate to **VPC Dashboard** → **VPCs**
- Click **Create VPC**
- Configurations:
 - Name: Test-VPC
 - IPv4 CIDR block: 192.168.1.0/24
 - IPv6 CIDR block: No IPv6 CIDR block
 - Tenancy: Default
- Click **Create VPC**

Enable DNS Hostname and DNS Resolution:



- Select your VPC (Test-VPC)
- Click **Actions** → **Edit VPC settings**
- Check both:
 - ✓ Enable DNS hostnames
 - ✓ Enable DNS resolution
- Click **Save**

Create Internet Gateway:

Internet gateway igw-011a59d43fa3790c0 successfully attached to vpc-0d775863b24659341

Notifications 0 0 2 0 0

igw-011a59d43fa3790c0 / Test-IGW

Actions

Details Info

Internet gateway ID
igw-011a59d43fa3790c0

State
Attached

VPC ID
vpc-0d775863b24659341 | Test-VPC

Owner
734846753465

Tags

Search tags

< 1 > ⚙

Key	Value
Name	Test-IGW

- Navigate to **VPC → Internet Gateways**
- Click **Create Internet Gateway**
- Name: **Test-IGW**
- Click **Create Internet gateway**
- Select the IGW → **Actions → Attach to VPC**
- Select **Test-VPC**
- Click **Attach Internet gateway**

Creating Public Subnet:

subnet-0f0658ac300a9a63b / Public-Subnet-NV

Actions

Details

Subnet ID
subnet-0f0658ac300a9a63b

IPv4 CIDR
192.168.1.0/25

Availability Zone
us-east-1a (us-east-1a)

Network ACL
acl-0f8e782e82b60ca43

Auto-assign customer-owned IPv4 address
No

IPv6 CIDR reservations
-

Resource name DNS AAAA record
Disabled

Subnet ARN
arn:aws:ec2:us-east-1:734846753465:subnet/subnet-0f0658ac300a9a63b

Available IPv4 addresses
123

Network border group
us-east-1

Default subnet
No

Customer-owned IPv4 pool
-

IPv6-only
No

DNS64
Disabled

State
Available

IPv6 CIDR
-

VPC
vpc-0d775863b24659341 | Test-VPC

Auto-assign public IPv4 address
Yes

Outpost ID
-

Hostname type
IP name

Owner
734846753465

Block Public Access
Off

IPv6 CIDR association ID
-

Route table
rtb-0706f7890265ee0c7

Auto-assign IPv6 address
No

IPv4 CIDR reservations
-

Resource name DNS A record
Disabled

- Navigate to **VPC → Subnets**
- Click **Create Subnet**
- Configuration:
 - ✓ VPC ID: **Test-VPC**
 - ✓ Subnet name: **Public-Subnet**
 - ✓ Availability Zone: **us-east-1a**
 - ✓ IPv4 CIDR block: **192.168.1.0/25**
- Click **Create Subnet**

Enable Auto-Assign Public IP for Public Subnet:

VPC > Subnets > subnet-0f0658ac300a9a63b > Edit subnet settings

Info

Subnet

Subnet ID

subnet-0f0658ac300a9a63b

Name

Public-Subnet-NV

Auto-assign IP settings

Info

Enable AWS to automatically assign a public IPv4 or IPv6 address to a new primary network interface for an instance in this subnet.

☒ Enable auto-assign public IPv4 address

Info

☐ Enable auto-assign customer-owned IPv4 address

Info

Option disabled because no customer owned pools found.

- Select **Public-Subnet-NV**
- Click **Actions** → **Edit subnet settings**
- Check: **Enable auto-assign public IPv4 address**
- Click **Save**

Create Public Route Table

rtb-06f374ed84be9a920 / Public-RT

Actions ▾

Details

Info

Route table ID

rtb-06f374ed84be9a920

VPC

vpc-0d775863b24659341 | Test-VPC

Main

No

Owner ID

734846753465

Explicit subnet associations

subnet-0f0658ac300a9a63b / Public-Subnet-NV

Edge associations

–

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (2)

Both ▾

Edit routes

Filter routes

< 1 > ⚙

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-011a59d43fa3790c0	Active	No	Create Route
192.168.1.0/24	local	Active	No	Create Route Table

- Navigate to **VPC** → **Route Tables**
- Click **Create Route Table**
- Name: **Public-RT**
- VPC: **Test-VPC**
- Click **Create Route Table**
- Select the RT → **Routes** tab
- Click **Edit Routes**
- Add route:
 - Destination: **0.0.0.0/0**
 - Target: **Internet Gateway** → **Test-IGW**
- **Save** and associate with Public-Subnet

Creating Private Subnet:

subnet-0cd6b8311a4442395 / Private-Subnet-NV

[Actions](#)

Details

Subnet ID[subnet-0cd6b8311a4442395](#)**IPv4 CIDR**[192.168.1.128/25](#)**Availability Zone**[use1-az4 \(us-east-1b\)](#)**Network ACL**[acl-0f8e782e82b60ca43](#)**Auto-assign customer-owned IPv4 address**

No

IPv6 CIDR reservations

-

Resource name DNS AAAA record

Disabled

Subnet ARN[arn:aws:ec2:us-east-1:734846753465:subnet/subnet-0cd6b8311a4442395](#)**Available IPv4 addresses**[123](#)**Network border group**[us-east-1](#)**Default subnet**

No

Customer-owned IPv4 pool

-

IPv6-only

No

DNS64

Disabled

State[Available](#)**IPv6 CIDR**

-

VPC[vpc-0d775863b24659341 | Test-VPC](#)**Auto-assign public IPv4 address**

Yes

Outpost ID

-

Hostname type

IP name

Owner[734846753465](#)**Block Public Access**[Off](#)**IPv6 CIDR association ID**

-

Route table[rtb-0706f7890265ee0c7](#)**Auto-assign IPv6 address**

No

IPv4 CIDR reservations

-

Resource name DNS A record

Disabled

- Navigate to **VPC → Subnets**
- Click **Create Subnet**
- Configuration:
 - ✓ VPC ID: **Test-VPC**
 - ✓ Subnet name: **Private-Subnet**
 - ✓ Availability Zone: **us-east-1b**
 - ✓ IPv4 CIDR block: **192.168.1.128/25**
- Click **Create Subnet**

Create Private Route Table

✓ You have successfully updated subnet associations for rtb-0bd4aaa96e3daa111 / Private-RT.



rtb-0bd4aaa96e3daa111 / Private-RT

[Actions](#)

Details [Info](#)

Route table ID[rtb-0bd4aaa96e3daa111](#)**VPC**[vpc-0d775863b24659341 | Test-VPC](#)**Main**[No](#)**Owner ID**[734846753465](#)**Explicit subnet associations**[subnet-0cd6b8311a4442395 / Private-Subnet-NV](#)**Edge associations**

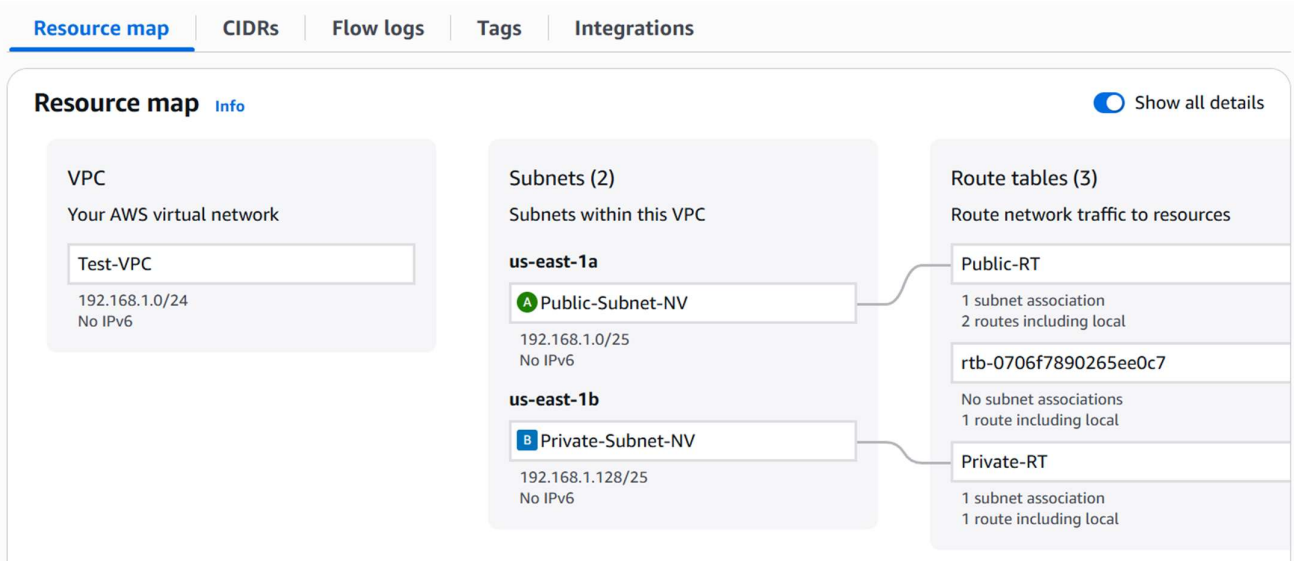
-

Routes[Subnet associations](#)[Edge associations](#)[Route propagation](#)[Tags](#)**Routes (1)**[Both](#)[Edit routes](#)[< 1 > ⚙](#)

Destination	Target	Status	Propagated	Route Origin
192.168.1.0/24	local	Active	No	Create Route Table

- Navigate to **VPC → Route Tables**
- Click **Create Route Table**
- Name: **Private-RT**
- VPC: **Test-VPC**
- Click **Create Route Table**
- Select the RT → **Subnet Associations** tab
- Click **Edit Subnet Associations**
- Select **Private-Subnet**
- **Save**

Verify the Setup:



Summary - Task 1 Complete:

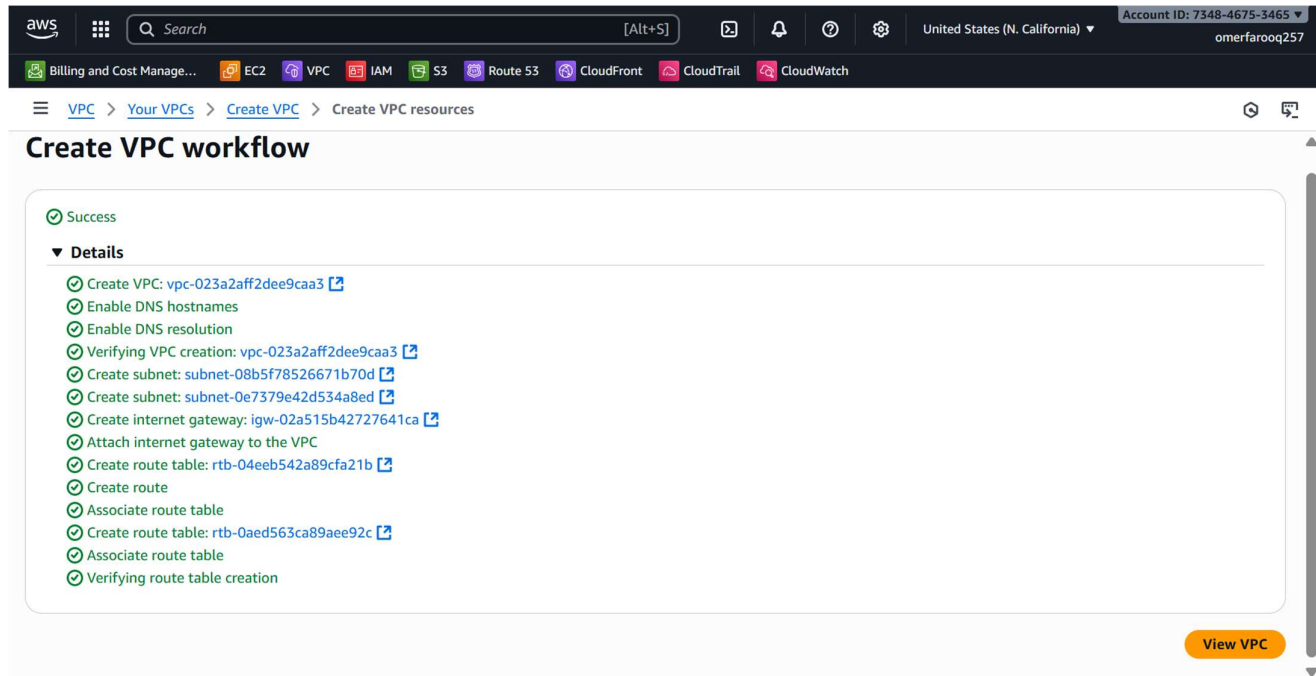
```
VPC: AdvancedVPC (192.168.1.0/24)
├─ PublicSubnet (192.168.1.0/25)
│   ├── IPs: 192.168.1.1 - 192.168.1.126
│   └─ Route Table: PublicRT
│       └─ Route: 0.0.0.0/0 → Internet Gateway
├─ PrivateSubnet (192.168.1.128/25)
│   ├── IPs: 192.168.1.129 - 192.168.1.254
│   └─ Route Table: PrivateRT
│       └─ Only local route (192.168.1.0/24)
└─ Internet Gateway: AdvancedIGW
    └─ DNS: Enabled
```

2. Enable VPC peering for cross-region.

Purpose: To enable private network connectivity between VPCs without internet exposure.

STEP TO ENABLE VPC PEERING FOR CROSS-REGION:

Create a Second VPC in a Different Region – N. California



Create VPC workflow

Success

Details

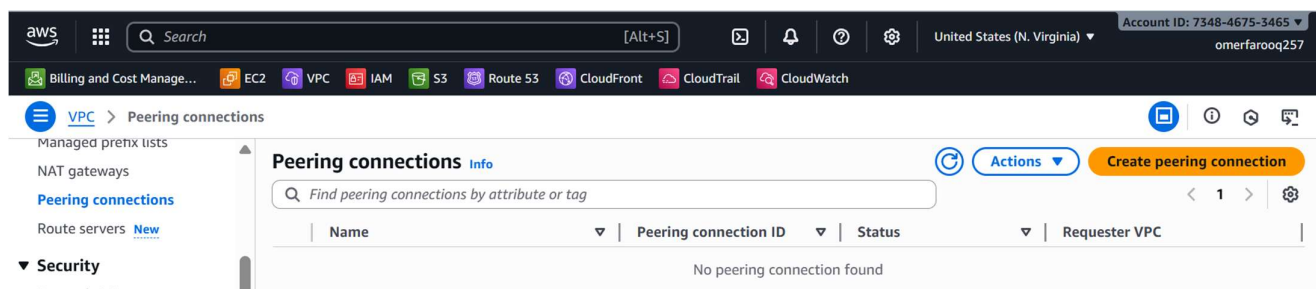
- ✓ Create VPC: vpc-023a2aff2dee9caa3
- ✓ Enable DNS hostnames
- ✓ Enable DNS resolution
- ✓ Verifying VPC creation: vpc-023a2aff2dee9caa3
- ✓ Create subnet: subnet-08b5f78526671b70d
- ✓ Create subnet: subnet-0e7379e42d534a8ed
- ✓ Create internet gateway: igw-02a515b42727641ca
- ✓ Attach internet gateway to the VPC
- ✓ Create route table: rtb-04eeb542a89cfa21b
- ✓ Create route
- ✓ Associate route table
- ✓ Create route table: rtb-0aed563ca89aee92c
- ✓ Associate route table
- ✓ Verifying route table creation

[View VPC](#)

- The architecture of the VPC is shown in the image below.



Create VPC Peering Connection (Requester Side)



Peering connections Info

[Actions](#) [Create peering connection](#)

Find peering connections by attribute or tag

Name	Peering connection ID	Status	Requester VPC
No peering connection found			

- Navigate to **VPC → Peering Connections**
- Click **Create Peering Connection**
- Configuration:
 - Peering connection name tag: Peering-NV-NC
 - **VPC ID (Requester):** vpc-0a1b2c3d4e5f6g7h8 (NV-VPC-1) - Region: us-east-1
 - VPC to accept peering: - Account: [Your AWS Account ID]
 - **VPC ID (Acceptor):** vpc-0i1j2k3l4m5n6o7p8 (NC-VPC) - Region: us-west-2
- Click **Create Peering Connection**

Create peering connection

A VPC peering connection is a networking connection between two VPCs that enables you to route traffic between them privately. [Info](#)

Peering connection settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

Peering-NV-NC

Select a local VPC to peer with

VPC ID (Requester)
vpc-08188c8ed1374447e (NV-VPC-1)

VPC CIDRs for vpc-08188c8ed1374447e (NV-VPC-1)

CIDR	Status	Status reason
192.168.1.0/24	Associated	-

Select another VPC to peer with

Account
☒ My account
☐ Another account

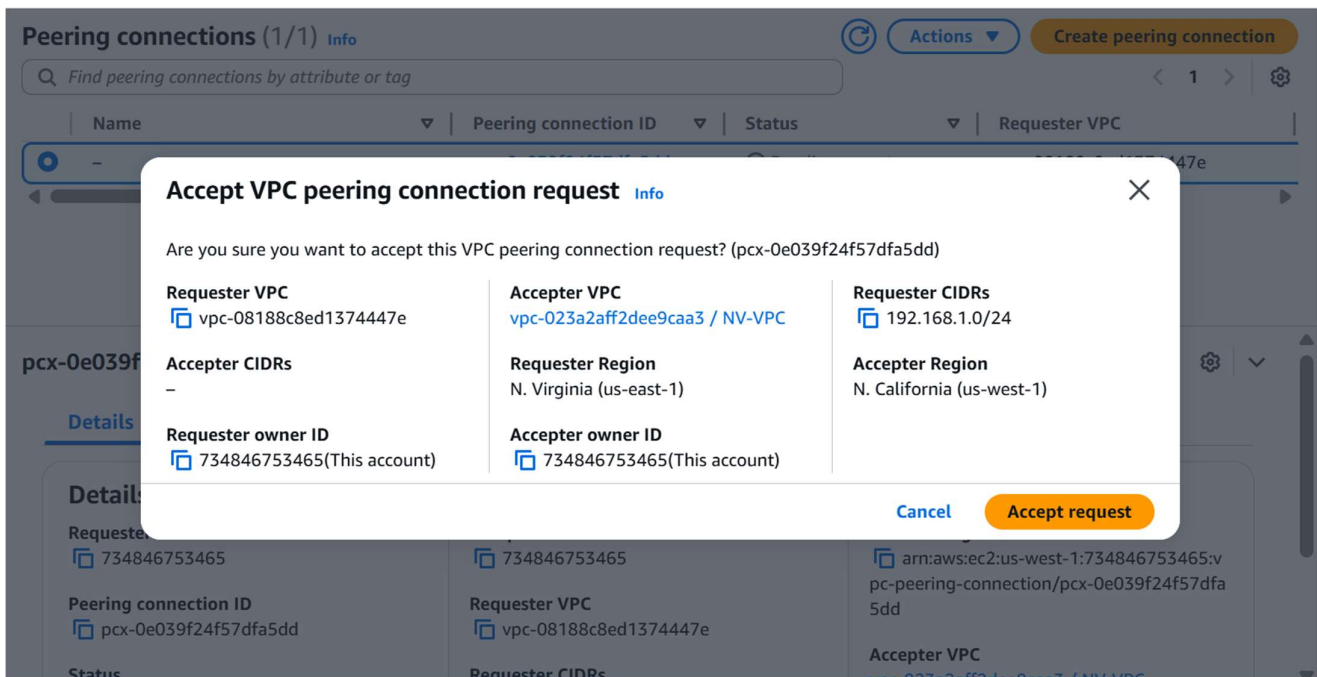
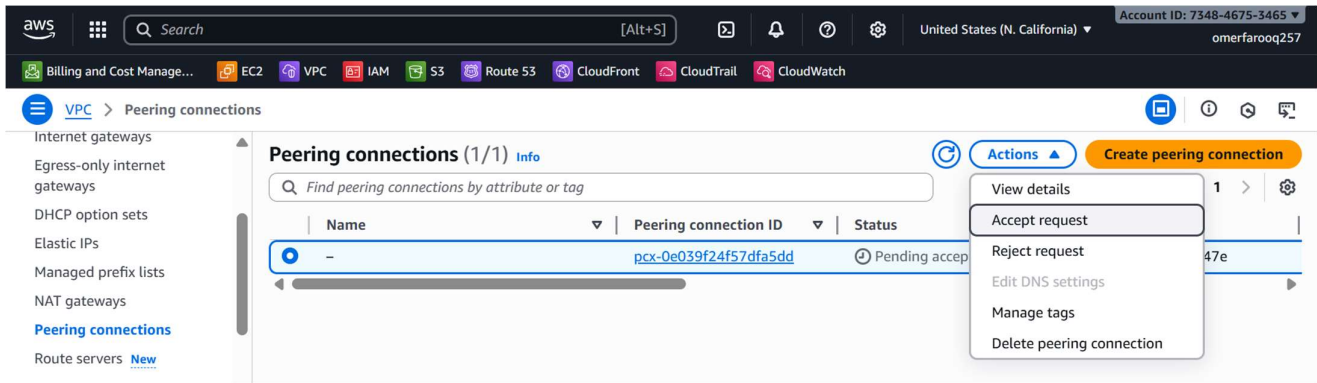
Region
☐ This Region (us-east-1)
☒ Another Region

United States (N. California) (us-west-1)

VPC ID (Acceptor)
vpc-023a2aff2dee9caa3

Accept VPC Peering Connection (Acceptor Side)

- Switch to the us-west-2 region in the AWS Console
- Navigate to **VPC → Peering Connections**
- Find the pending peering connection
- Select it. **Actions → Accept Request**
- Click **Accept Peering Connection**



Enable DNS Resolution Over Peering

- Navigate to **VPC → Peering Connections**
- Select the peering connection
- Click **Actions → Edit Peering Connection Options**
- Check: ☒ **Allow DNS resolution from remote VPC endpoint**
- Click **Save**

Update Route Tables (Requester Side - us-east-1)

- Navigate to **VPC → Route Tables**
- Select **NV-Public-RT**
- **Routes tab → Edit Routes**

Destination	Target	Status	Propagated	Route Origin	
10.0.0.0/16	local	Active	No	CreateRouteTable	
192.168.1.0/24	Peering Connection	Active	No	CreateRoute	Remove
0.0.0.0/0	Internet Gateway	Active	No	CreateRoute	Remove

Buttons: Add route, Cancel, Preview, Save changes

- Click **Add Route**
 - Destination: **10.0.0.0/16**
 - Target: **Peering Connection → Peering-NV-NC**
- Save
- Repeat for NV-Private-RT

Destination: 10.0.0.0/16 | Target: Peering Connection | Status: No | Origin: CreateRoute | Remove

Buttons: Add route, Cancel, Preview, Save changes

Update Route Tables (Accepter Side - us-west-2)

- Switch to us-west-2 region
- Navigate to **VPC → Route Tables**
- Select the west subnet's route table
- Edit Routes
- Add route:
 - Destination: **192.168.1.0/24**
 - Target: **Peering Connection → East-West-Peering**

aws [Search] [Alt+S] United States (N. California) Account ID: 7348-4675-3465 omerfaroq257

Billing and Cost Manage... EC2 VPC IAM S3 Route 53 CloudFront CloudTrail CloudWatch

VPC > Route tables > rtb-0f9e89de77f6ee86e > Edit routes

Edit routes

Destination	Target	Status	Propagated	Route Origin	
10.0.0.0/16	local	Active	No	CreateRouteTable	
0.0.0.0/0	Internet Gateway	Active	No	CreateRoute	Remove
192.168.1.0/24	Peering Connection	-	No	CreateRoute	Remove

Add route

Cancel Preview Save changes

Update Security Groups for Peering for instances in NV-VPC (us-east-1):

- Click on **Add rule**
 - Protocol: **All ICMP – IPv4**
 - Custom CIDR: **10.0.0.0/16**

Update Security Groups for Peering for instances in NC-VPC (us-west-1):

- Click on **Add rule**
 - Protocol: **All ICMP – IPv4**
 - Custom CIDR: **192.168.1.0/16**

aws [Search] [Alt+S] United States (N. Virginia) Account ID: 7348-4675-3465 omerfaroq257

Billing and Cost Manage... EC2 VPC IAM S3 Route 53 CloudFront CloudTrail CloudWatch

EC2 > Security Groups > sg-0c8d892bbc05b7f15 - SG-VPC-1-NV > Edit inbound rules

Security Group ID	Protocol	Port Range	Source	Action
sgr-02a659b9161b4a3d3	SSH	TCP 22	192.168.5.0/24	Delete
sgr-0cabfb873fd27a23e	HTTP	TCP 80	0.0.0.0/0	Delete
sgr-03f046e8b2caf4b09	All ICMP - IPv4	ICMP All	0.0.0.0/0	Delete
sgr-0522303174069211d	All ICMP - IPv4	ICMP All	192.168.2.0/24	Delete
sgr-0c419f99ebae2ac7b	All ICMP - IPv4	ICMP All	192.168.4.0/24	Delete
			10.0.0.0/16	

Add rule

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Cancel Preview changes Save rules

aws [Search] [Alt+S] United States (N. California) Account ID: 7348-4675-3465 omerfaroq257

Billing and Cost Manage... EC2 VPC IAM S3 Route 53 CloudFront CloudTrail CloudWatch

EC2 > Security Groups > sg-0acb69b17219b70e1 - SG-VPC-NC-PUB > Edit inbound rules

Edit inbound rules Info

Inbound rules control the incoming traffic that's allowed to reach the instance.

Security group rule ID	Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info	
sgr-0c67e081dc4ddc92c	HTTP	TCP	80	Cu... 0.0.0.0/0		Delete
sgr-05f62e641fcd1c972	SSH	TCP	22	Cu... 0.0.0.0/0		Delete
sgr-0b1c64405c6e2ee14	All ICMP - IPv4	ICMP	All	Cu... 192.168.1.0/24		Delete

[Add rule](#)

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

[Cancel](#) [Preview changes](#) [Save rules](#)

Test the connection between two VPCs in different regions:

- Launch an EC2 instance in N. Virginia and another in N. California.
- Configure the instances and assign the appropriate:
 - VPC
 - Subnet
 - Key
 - Security Group
- Launch the instances in Git Bash and check the connectivity between the two servers.
 - # ping 10.0.3.184
 - # ping 192.168.1.8

```

root@ip-192-168-1-8:/home/ec2-user
.1 ms
64 bytes from 10.0.3.184: icmp_seq=174 ttl=127 time=62
.1 ms
64 bytes from 10.0.3.184: icmp_seq=175 ttl=127 time=62
.1 ms
64 bytes from 10.0.3.184: icmp_seq=176 ttl=127 time=62
.1 ms
64 bytes from 10.0.3.184: icmp_seq=177 ttl=127 time=62
.1 ms
64 bytes from 10.0.3.184: icmp_seq=178 ttl=127 time=62
.1 ms
64 bytes from 10.0.3.184: icmp_seq=179 ttl=127 time=62
.1 ms
64 bytes from 10.0.3.184: icmp_seq=180 ttl=127 time=62
.1 ms
64 bytes from 10.0.3.184: icmp_seq=181 ttl=127 time=62
.1 ms
64 bytes from 10.0.3.184: icmp_seq=182 ttl=127 time=62
.1 ms
64 bytes from 10.0.3.184: icmp_seq=183 ttl=127 time=62
.1 ms
64 bytes from 10.0.3.184: icmp_seq=184 ttl=127 time=62
.1 ms
64 bytes from 10.0.3.184: icmp_seq=185 ttl=127 time=62
.2 ms
64 bytes from 10.0.3.184: icmp_seq=186 ttl=127 time=62
.2 ms
64 bytes from 10.0.3.184: icmp_seq=187 ttl=127 time=62
.1 ms
64 bytes from 10.0.3.184: icmp_seq=188 ttl=127 time=62
.1 ms
64 bytes from 10.0.3.184: icmp_seq=189 ttl=127 time=62
.1 ms

root@ip-10-0-3-184:/home/ec2-user
2.1 ms
64 bytes from 192.168.1.8: icmp_seq=168 ttl=127 time=6
2.0 ms
64 bytes from 192.168.1.8: icmp_seq=169 ttl=127 time=6
2.1 ms
64 bytes from 192.168.1.8: icmp_seq=170 ttl=127 time=6
2.1 ms
64 bytes from 192.168.1.8: icmp_seq=171 ttl=127 time=6
2.1 ms
64 bytes from 192.168.1.8: icmp_seq=172 ttl=127 time=6
2.0 ms
64 bytes from 192.168.1.8: icmp_seq=173 ttl=127 time=6
2.0 ms
64 bytes from 192.168.1.8: icmp_seq=174 ttl=127 time=6
2.0 ms
64 bytes from 192.168.1.8: icmp_seq=175 ttl=127 time=6
2.3 ms
64 bytes from 192.168.1.8: icmp_seq=176 ttl=127 time=6
2.2 ms
64 bytes from 192.168.1.8: icmp_seq=177 ttl=127 time=6
2.0 ms
64 bytes from 192.168.1.8: icmp_seq=178 ttl=127 time=6
2.2 ms
64 bytes from 192.168.1.8: icmp_seq=179 ttl=127 time=6
2.1 ms
64 bytes from 192.168.1.8: icmp_seq=180 ttl=127 time=6
2.2 ms
64 bytes from 192.168.1.8: icmp_seq=181 ttl=127 time=6
2.1 ms
64 bytes from 192.168.1.8: icmp_seq=182 ttl=127 time=6
2.1 ms
64 bytes from 192.168.1.8: icmp_seq=183 ttl=127 time=6
2.1 ms

```

3. **Enable VPC peering for cross-account. (You can collaborate with your friend and do this task.)**

4. Set up a VPC Transit gateway.

Purpose: To simplify VPC-to-VPC and VPC-to-on-premises connectivity. Alternative to VPC Peering, and when planning to add more VPCs. Easier than managing multiple peering connections.

STEPS TO CREATE A TRANSIT GATEWAY:

Create 4 VPCs in any region. In this task, I have created four VPCs in the North Virginia region.

☰ [VPC](#) > [Your VPCs](#) > [Create VPC](#) > Create VPC resources

Create VPC workflow

✔ Success

▼ Details

- ✔ Create VPC: [vpc-0a86e53d4070a0eab](#) ↗
- ✔ Enable DNS hostnames
- ✔ Enable DNS resolution
- ✔ Verifying VPC creation: [vpc-0a86e53d4070a0eab](#) ↗
- ✔ Create subnet: [subnet-0d51af1d84f031e69](#) ↗
- ✔ Create subnet: [subnet-039282539dfa6808b](#) ↗
- ✔ Create internet gateway: [igw-0a4caf4f610708850](#) ↗
- ✔ Attach internet gateway to the VPC
- ✔ Create route table: [rtb-0d1f7d5e2b54059d4](#) ↗
- ✔ Create route
- ✔ Associate route table
- ✔ Create route table: [rtb-0aced09586e68777b](#) ↗
- ✔ Associate route table
- ✔ Verifying route table creation

The resource map of all four VPCs is attached below.

Your VPCs (1/4) Info									
Find VPCs by attribute or tag									
Last updated 1 minute ago Actions Create VPC									
< 1 > ⚙									
☐	Name	VPC ID	State	Bloc...	IPv4 CIDR	IPv6 ...	DHCP option set	Main route table	
☐	NV-VPC-3	vpc-02f23d7c9c7b507a8	Available	Off	192.168.3.0/24	-	dopt-09a84d39e5d39a...	rtb-06c6512b6d32aaa5	
☐	NV-VPC-4	vpc-001ca49c73371a1e4	Available	Off	192.168.4.0/24	-	dopt-09a84d39e5d39a...	rtb-0531cb52a744a216	
☒	NV-VPC-1	vpc-08188c8ed1374447e	Available	Off	192.168.1.0/24	-	dopt-09a84d39e5d39a...	rtb-074e91d93e5c9409	
☐	NV-VPC-2	vpc-0a86e53d4070a0eab	Available	Off	192.168.2.0/24	-	dopt-09a84d39e5d39a...	rtb-01b43698ff997eb2f	

vpc-08188c8ed1374447e / NV-VPC-1

[Details](#) [Resource map](#) [CIDRs](#) [Flow logs](#) [Tags](#) [Integrations](#)

Resource map [Info](#)

☐ Show all details

VPC

Your AWS virtual network

NV-VPC-1

Subnets (2)

Subnets within this VPC

us-east-1a

NV-Public-SN-1

NV-Private-SN-1

Route tables (3)

Route network traffic to resources

NV-Public-RT-1

NV-Private-RT-1

[rtb-074e91d93e5c94097](#)

Network Connect

Connections to oth

NV-IGW-1

Your VPCs (1/4) [Info](#)

Last updated 2 minutes ago [Actions](#) [Create VPC](#)

☐	Name	VPC ID	State	Bloc...	IPv4 CIDR	IPv6 ...	DHCP option set	Main route table	
☐	NV-VPC-3	vpc-02f23d7c9c7b507a8	Available	Off	192.168.3.0/24	-	dopt-09a84d39e5d39a...	rtb-06c6512b6d32aaa5	
☐	NV-VPC-4	vpc-001ca49c73371a1e4	Available	Off	192.168.4.0/24	-	dopt-09a84d39e5d39a...	rtb-0531cb52a744a216	
☐	NV-VPC-1	vpc-08188c8ed1374447e	Available	Off	192.168.1.0/24	-	dopt-09a84d39e5d39a...	rtb-074e91d93e5c9409	
☒	NV-VPC-2 ✎	vpc-0a86e53d4070a0eab	Available	Off	192.168.2.0/24	-	dopt-09a84d39e5d39a...	rtb-01b43698ff997eb2f	

vpc-0a86e53d4070a0eab / NV-VPC-2

[Details](#) [Resource map](#) [CIDRs](#) [Flow logs](#) [Tags](#) [Integrations](#)

Resource map [Info](#)

☐ Show all details

VPC

Your AWS virtual network

NV-VPC-2

Subnets (2)

Subnets within this VPC

us-east-1a

NV-Public-SN-2

NV-Private-SN-2

Route tables (3)

Route network traffic to resources

NV-Private-RT-2

NV-Public-RT-2

[rtb-01b43698ff997eb26](#)

Network Connect

Connections to oth

NV-IGW-2

Your VPCs (1/4) [Info](#) Last updated 2 minutes ago [Actions](#) [Create VPC](#)

Find VPCs by attribute or tag

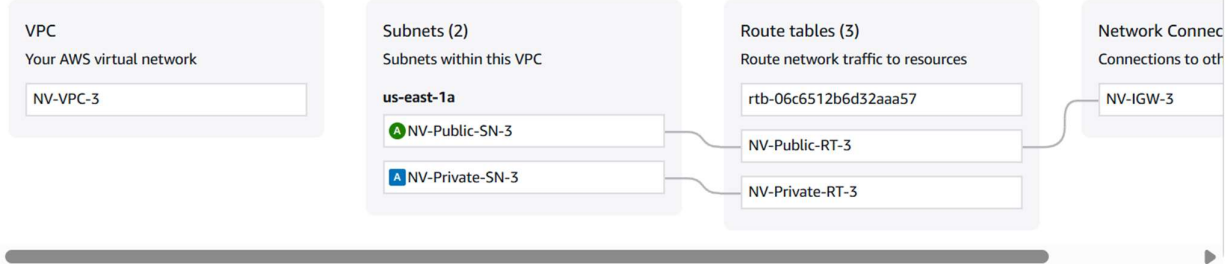
	Name	VPC ID	State	Bloc...	IPv4 CIDR	IPv6 ...	DHCP option set	Main route table
☒	NV-VPC-3	vpc-02f23d7c9c7b507a8	Available	Off	192.168.3.0/24	-	dopt-09a84d39e5d39a...	rtb-06c6512b6d32aaa5
☐	NV-VPC-4	vpc-001ca49c73371a1e4	Available	Off	192.168.4.0/24	-	dopt-09a84d39e5d39a...	rtb-0531cb52a744a216
☐	NV-VPC-1	vpc-08188c8ed1374447e	Available	Off	192.168.1.0/24	-	dopt-09a84d39e5d39a...	rtb-074e91d93e5c9409
☐	NV-VPC-2	vpc-0a86e53d4070a0eab	Available	Off	192.168.2.0/24	-	dopt-09a84d39e5d39a...	rtb-01b43698ff997eb2f

vpc-02f23d7c9c7b507a8 / NV-VPC-3

[Details](#) [Resource map](#) [CIDRs](#) [Flow logs](#) [Tags](#) [Integrations](#)

Resource map [Info](#)

☐ Show all details



Your VPCs (1/4) [Info](#)

Last updated 3 minutes ago

[Actions](#)

[Create VPC](#)

Find VPCs by attribute or tag

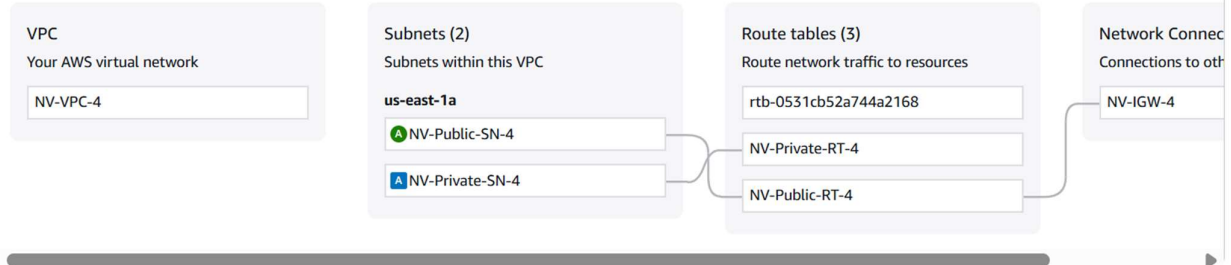
	Name	VPC ID	State	Bloc...	IPv4 CIDR	IPv6 ...	DHCP option set	Main route table
☐	NV-VPC-3	vpc-02f23d7c9c7b507a8	Available	Off	192.168.3.0/24	-	dopt-09a84d39e5d39a...	rtb-06c6512b6d32aaa5
☒	NV-VPC-4	vpc-001ca49c73371a1e4	Available	Off	192.168.4.0/24	-	dopt-09a84d39e5d39a...	rtb-0531cb52a744a216
☐	NV-VPC-1	vpc-08188c8ed1374447e	Available	Off	192.168.1.0/24	-	dopt-09a84d39e5d39a...	rtb-074e91d93e5c9409
☐	NV-VPC-2	vpc-0a86e53d4070a0eab	Available	Off	192.168.2.0/24	-	dopt-09a84d39e5d39a...	rtb-01b43698ff997eb2f

vpc-001ca49c73371a1e4 / NV-VPC-4

[Details](#) [Resource map](#) [CIDRs](#) [Flow logs](#) [Tags](#) [Integrations](#)

Resource map [Info](#)

☐ Show all details



Creating a Transit Gateway:

- Region: United States (N. Virginia)
- Navigate to **VPC** → **Transit Gateway**
- Click on **Create transit gateway**

- **Configuration:**
 - DNS support: ✓ Enable
 - VPN ECMP support: ✓ Enable
 - Default route table association: ✓ Enable
 - Default route table propagation: ✓ Enable

Create transit gateway [Info](#)

A transit gateway (TGW) is a network transit hub that interconnects attachments (VPCs and VPNs) within the same AWS account or across AWS accounts.

Details - optional

Name tag
Creates a tag with the key set to Name and the value set to the specified string.

NV-TGW

Description [Info](#)
Set the description of your transit gateway to help you identify it in the future.

Routing and connecting all the VPCs

Configure the transit gateway

Amazon side Autonomous System Number (ASN) [Info](#)

ASN

☒ DNS support [Info](#)

☐ Security Group Referencing support [Info](#)

☒ VPN ECMP support [Info](#)

☒ Default route table association [Info](#)

☒ Default route table propagation [Info](#)

☐ Multicast support [Info](#)

- Click "Create transit gateway"
- Wait 5-10 minutes for state: "available"
- Note Transit Gateway ID: tgw-california-xxxxx

✓ You successfully created tgw-033e87181d8a4b05d / NV-TGW. ✕

Transit gateways (1/1) [Info](#) 🔄 Actions Create transit gateway

🔍 Find transit gateway by attribute or tag < 1 > ⚙️

☒	Name 🔗	Transit gateway ID	Owner ID	State
☒	NV-TGW 🔗	tgw-033e87181d8a4b05d	734846753465	⌛ Pending

Create VPC Attachments:

- VPC Dashboard → "Transit gateway attachments"
- Click "Create transit gateway attachment"
- Configure Attachment
 - Name tag: **TGA-VPC-1**
 - Transit gateway ID: **tgw-0165d3dcd7f9a2c63**
 - Attachment type: **VPC**
 - DNS support: ✓ **Enable**
 - VPC ID: **NV-VPC-1 (192.168.1.0/24)**

The screenshot shows the AWS Management Console interface for creating a transit gateway attachment. The top navigation bar includes the AWS logo, a search bar, and various service icons. The breadcrumb trail indicates the path: VPC > Transit gateway attachments > Create transit gateway attachment. A descriptive text states: "A transit gateway (TGW) is a network transit hub that interconnects attachments (VPCs and VPNs) within the same AWS account or across AWS accounts."

The main configuration area is divided into two sections:

- Details:** This section contains three input fields:
 - Name tag - optional:** A text input field with the value "TGA-VPC-2". A tooltip explains: "Creates a tag with the key set to Name and the value set to the specified string."
 - Transit gateway ID:** A dropdown menu with the selected value "tgw-0165d3dcd7f9a2c63".
 - Attachment type:** A dropdown menu with the selected value "VPC".
- VPC attachment:** This section features a search bar and a list of available VPCs for selection:

VPC ID	IP Range
vpc-02f23d7c9c7b507a8 (NV-VPC-3)	192.168.3.0/24
vpc-001ca49c73371a1e4 (NV-VPC-4)	192.168.4.0/24
vpc-08188c8ed1374447e (NV-VPC-1)	192.168.1.0/24
vpc-0a86e53d4070a0eab (NV-VPC-2)	192.168.2.0/24

Below the list is a "Select a VPC" dropdown menu.

The footer of the console shows "CloudShell", "Feedback", and copyright information for Amazon Web Services, Inc. or its affiliates, along with links for "Privacy", "Terms", and "Cookie preferences".

- Repeat the same steps for the remaining VPCs to create transit gateway attachments.
- Configure the attachment with the appropriate VPCs.

Configure the route tables for each VPC.

- Navigate to "Route tables"
- Select **NV-Public-RT-1** & **NV-Private-RT-1**
- "Routes" → "Edit routes"
- Configure:
 - Destination: 192.168.2.0/24 Target: Transit Gateway → TGW-NV (VPC-2)
 - Destination: 192.168.3.0/24 Target: Transit Gateway → TGW-NV (VPC-3)
 - Destination: 192.168.4.0/24 Target: Transit Gateway → TGW-NV (VPC-4)

Edit routes

Destination	Target	Status	Propagated	Route Origin	
192.168.1.0/24	local	Active	No	CreateRouteTable	
0.0.0.0/0	Internet Gateway	Active	No	CreateRoute	Remove
192.168.2.0/24	Transit Gateway	-	No	CreateRoute	Remove
192.168.3.0/24	Transit Gateway	-	No	CreateRoute	Remove
192.168.4.0/24	Transit Gateway	-	No	CreateRoute	Remove

[Add route](#) [Cancel](#) [Preview](#) [Save changes](#)

- Repeat the above steps for the remaining VPCs' Route Tables.
- Make sure to use the appropriate CIDR Range.

Launch EC2 Instances and Update Security Groups:

Instances (1/4) [Info](#) Last updated 7 minutes ago [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

[All states](#)

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status
☒	Server-VPC-1	i-0be390587404cb734	Running	t2.micro	2/2 checks passed	View alarms
☐	Server-VPC-4	i-05fadde43902accc1	Running	t2.micro	2/2 checks passed	View alarms
☐	Server-VPC-2	i-09eb5f5cfbcd6bd44	Running	t2.micro	2/2 checks passed	View alarms
☐	Server-VPC-3	i-014e40d7317214bd5	Running	t2.micro	2/2 checks passed	View alarms

- EC2 → Instances → **Launch Instance**
- Basic configurations:
 - Name: Server-VPC-1
 - Key pair: key.pem
 - VPC – NV-VPC-1
 - Subnet: NV-Public-SN-1

EC2 > Instances > Launch an instance

Network settings Info

VPC - required Info

vpc-08188c8ed1374447e (NV-VPC-1)
192.168.1.0/24

Subnet Info

subnet-0cd58de93da011e3a NV-Public-SN-1
VPC: vpc-08188c8ed1374447e Owner: 734846753465
Availability Zone: us-east-1a (use1-az2) Zone type: Availability Zone
IP addresses available: 11 CIDR: 192.168.1.0/28

Auto-assign public IP Info

Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group Select existing security group

Security group name - required

SG-VPC-1-NV

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and _-:/()#,@!+=8;()!\$*

Description - required Info

Security Group of VPC 1

Summary

Number of instances Info

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.9.2...read more
ami-052064a798f08f0d3

Virtual server type (instance type)

t2.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet. Data transfer charges are not included as part of the free tier allowance.

Cancel Launch instance Preview code

- Configure the **Network Settings → Firewall (Security Groups)** as shown below and save the security group for future use.
- We can create the Security Groups separately, but to reduce a step, I have created and configured them here while launching an instance.

EC2 > Instances > Launch an instance

Security group rule 2 (ICMP, All, 192.168.2.0/24) Remove

Type Info

All ICMP - IPv4

Protocol Info

ICMP

Port range Info

All

Source type Info

Custom

Source Info

Add CIDR, prefix list or security group

192.168.2.0/24 X

Description - optional Info

e.g. SSH for admin desktop

Security group rule 3 (ICMP, All, 192.168.3.0/24) Remove

Type Info

Custom ICMP - IPv4

Protocol Info

All

Port range Info

All

Source type Info

Custom

Source Info

Add CIDR, prefix list or security group

192.168.3.0/24 X

Description - optional Info

e.g. SSH for admin desktop

Security group rule 4 (ICMP, All, 192.168.4.0/24) Remove

Type Info

Custom ICMP - IPv4

Protocol Info

All

Port range Info

All

Source type Info

Custom

Source Info

Add CIDR, prefix list or security group

192.168.4.0/24 X

Description - optional Info

e.g. SSH for admin desktop

Summary

Number of instances Info

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.9.2...read more
ami-052064a798f08f0d3

Virtual server type (instance type)

t2.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet. Data transfer charges are not included as part of the free tier allowance.

Cancel Launch instance Preview code

- Repeat the steps and launch a total of four instances, one in each VPC's public subnet.
- Once all the configurations are done, test the connectivity by pinging each other.
- If everything goes right, the output will look like the image attached below.

The image displays four terminal windows, each showing the output of a ping command from a different EC2 instance. The windows are arranged in a 2x2 grid. Each window shows a series of ping results, including the time taken for the packet to reach the destination and return, the size of the packet, and the sequence number of the ICMP echo request. The destinations are IP addresses in the 192.168.0.0/16 range, which are likely the private IP addresses of the other three EC2 instances.

Terminal Window	Host	Destination IP	Time (ms)	Size (bytes)	ICMP Seq	TTL	Time=0
Top Left	192-168-3-5	192.168.4.7	.772, .787, .784, .770, .752, .856, .912, .814, .850	64	129-136	126	time=0
Top Right	192-168-1-8	192.168.2.12	0.786, 0.781, 0.835, 0.835, 0.738, 0.773, 0.728, 0.744, 0.869	64	287-294	126	time=
Bottom Left	192-168-2-12	192.168.3.5	.809, .803, .706, .794, .740, .837, .702, .823, .731	64	197-204	126	time=0
Bottom Right	192-168-4-7	192.168.1.8	752, 867, 798, 840, 699, 747, 759, 788, 819	64	64-71	126	time=0.

5. Set up VPC End Point.

Purpose: To access AWS services privately, avoiding internet exposure and reducing data transfer costs. Eliminate NAT Gateway charges for AWS services and keep traffic within the AWS network.

STEPS TO CREATE AND SET UP A VPC ENDPOINT:

Create a VPC, Subnets (Public & Private), a Routing Table, and an Internet Gateway.

vpc-0e74d84df4d37625e / VPC-NV

Actions

Details

VPC ID
vpc-0e74d84df4d37625e

DNS resolution
Enabled

Main network ACL
acl-0f54c38abcebea42e

IPv6 CIDR (Network border group)
-

State
Available

Tenancy
default

Default VPC
No

Network Address Usage metrics
Disabled

Block Public Access
Off

DHCP option set
dopt-09a84d39e5d39ac97

IPv4 CIDR
192.168.1.0/24

Route 53 Resolver DNS Firewall rule groups
-

DNS hostnames
Enabled

Main route table
rtb-079d55ea3c6b084b9

IPv6 pool
-

Owner ID
734846753465

Resource map

CIDRs

Flow logs

Tags

Integrations

Resource map

VPC

Your AWS virtual network

VPC-NV

Subnets (2)

Subnets within this VPC

us-east-1a

Public-SN

Private-SN

Route tables (3)

Route network traffic to resources

Private-RT

rtb-079d55ea3c6b084b9

Public-RT

Network Connections (1)

Connections to other network

IGW-NV

Create two Instances – a Public and a Private.

aws Search [Alt+S] United States (N. Virginia) Account ID: 7348-4675-3465 omerfarooq257

Billing and Cost Manage... EC2 VPC IAM S3 Route 53 CloudFront CloudTrail CloudWatch

EC2 > Instances

Instances (2) Info Last updated less than a minute ago Connect Instance state Actions Launch instances

Find Instance by attribute or tag (case-sensitive) All states

	Name	Instance ID	Instance state	Instanc...	Status check	Alarm status	Availability Zone	Public IPv4 DNS
☐	Public-Server	i-0b1a1f3b...	Running	t2.micro	2/2 checks pass	View alarms +	us-east-1a	ec2-44-211-193-154.co...
☐	Private-Server	i-07d41c7a...	Running	t2.micro	2/2 checks pass	View alarms +	us-east-1a	-

Create an S3 bucket.

Amazon S3 > Buckets

✔ Successfully created bucket "s3-endpoint-bucket-nv-test"
To upload files and folders, or to configure additional bucket settings, choose [View details](#).

[General purpose buckets](#) [All AWS Regions](#) [Directory buckets](#)

General purpose buckets (1) [Info](#)

[Refresh](#) [Copy ARN](#) [Empty](#) [Delete](#) [Create bucket](#)

Buckets are containers for data stored in S3.

Find buckets by name

Name	AWS Region	Creation date
s3-endpoint-bucket-nv-test	US East (N. Virginia) us-east-1	October 13, 2025, 00:04:58 (UTC+05:30)

Generate Access keys to configure AWS CLI in the Instances.

- Navigate to IAM → Access Management → Access keys → Create access key

aws [Search] [Alt+S] Account ID: 7348-4675-3465 omerfarooq257

Billing and Cost Manage... EC2 VPC IAM S3 Route S3 CloudFront CloudTrail CloudWatch

IAM > Security credentials

Identity and Access Management (IAM)

Search IAM

Dashboard

▼ **Access management**

- User groups
- Users
- Roles

Access keys (0) [Create access key](#)

Use access keys to send programmatic calls to AWS from the AWS CLI, AWS Tools for PowerShell, AWS SDKs, or direct AWS API calls. You can have a maximum of two access keys (active or inactive) at a time. [Learn more](#)

Access key ID	Created on	Access key last used	Region last used	Service last used	Status
No access keys					

As a best practice, avoid using long-term credentials like access keys. Instead, use tools which provide short term credentials. [Learn more](#)

[Create access key](#)

```

root@ip-192-168-1-11:/home/ec2-user
[root@ip-192-168-1-11 ec2-user]# aws --version
aws-cli/2.30.4 Python/3.9.23 Linux/6.1.153-175.280.amzn2023.x86_64 source/x86_64
.amzn.2023
[root@ip-192-168-1-11 ec2-user]# aws s3 ls

Unable to locate credentials. You can configure credentials by running "aws conf
figure".
[root@ip-192-168-1-11 ec2-user]# aws configure
AWS Access Key ID [None]: AKIA2WGCHTK43ZQ5NX6G
AWS Secret Access Key [None]: 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
Default region name [None]: us-east-1
Default output format [None]: json
[root@ip-192-168-1-11 ec2-user]# aws s3 ls
2025-10-12 18:34:58 s3-endpoint-bucket-nv-test
[root@ip-192-168-1-11 ec2-user]# |

```

- Access **Private Instance** using **Bastion/Jump Server**.
- Configure AWS CLI in the Private Instance.

```

[root@ip-192-168-1-11 ec2-user]# vi test.pem
[root@ip-192-168-1-11 ec2-user]# chmod 400 test.pem
[root@ip-192-168-1-11 ec2-user]# ssh -i test.pem ec2-user@192.168.1.134
The authenticity of host '192.168.1.134 (192.168.1.134)' can't be established.
ED25519 key fingerprint is SHA256:lvSGoITXYMVkpRckFLEI1tBo2/SIhpy48Wf1Ka8mkFI.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.1.134' (ED25519) to the list of known hosts.

#_
~\##### Amazon Linux 2023
~~\#####\
~~\###|
~~\#|
~~V~'-'> https://aws.amazon.com/linux/amazon-linux-2023
~~~
~~~.~.~
~~~\~\~
~/_/m/'~\~\~

[ec2-user@ip-192-168-1-134 ~]$ sudo su
[root@ip-192-168-1-134 ec2-user]# aws s3 ls

Unable to locate credentials. You can configure credentials by running "aws configure".
[root@ip-192-168-1-134 ec2-user]# aws configure
AWS Access Key ID [None]: AKIA2WGCHTK43ZQ5NX6G
AWS Secret Access Key [None]: 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
Default region name [None]: us-east-1
Default output format [None]: json
[root@ip-192-168-1-134 ec2-user]# aws s3 ls

```

Creating Endpoint:

- Navigate to **VPC → Endpoints**
- Click **Create Endpoint**
- **Configuration:**
 - **Name tag:** S3-Endpoint
 - **Service category:** AWS services
 - **Service name:** com.amazonaws.us-east-1.s3
 - **VPC:** NV-VPC (vpc-0a1b2c3d4e5f6g7h8)
 - **Route tables:**
 - ☒ Public-RT (rtb-0x1y2z3a4b5c6d7e8)
 - ☒ Private-RT (rtb-0m1n2o3p4q5r6s7t8)
 - **Policy:** Full Access (or custom)

The screenshot displays the AWS Management Console interface. The top navigation bar shows the AWS logo, a search bar, and various service icons. The main content area is divided into two sections: 'Endpoints' and 'Create endpoint'.

Endpoints Section:

- PrivateLink and Lattice:** A sidebar menu with options: Getting started, Endpoints (selected), Endpoint services, Service networks, and Lattice services.
- Endpoints Info:** A search bar with the placeholder text 'Find endpoints by attribute or tag'. Below it is a table with columns: Name, VPC endpoint ID, Endpoint type, and Status. The table is currently empty, displaying 'No endpoint found'.
- Actions:** A button labeled 'Create endpoint'.

Create endpoint Section:

- Name tag - optional:** A text input field with the value 'S3-Endpoint'.
- Type:** A section titled 'Select a category' with several radio button options:
 - AWS services:** Connect to services provided by Amazon with an interface endpoint, or a Gateway endpoint. (Selected)
 - EC2 Instance Connect Endpoint:** An elastic network interface that allows you to connect to resources in a private subnet.
 - Endpoint services that use NLBs and GWLBs:** Find services shared with you by service name. Connect to a Network LoadBalancer (NLB) service with an interface endpoint or to a Gateway LoadBalancer (GWLB) service with a Gateway Load Balancer endpoint.
 - PrivateLink Ready partner services:** Connect to SaaS services which have AWS Service Ready designation with an interface endpoint. Uses AWS PrivateLink.
 - Resources:** Connect to resources like Amazon Relational Database Services (RDS) with a Resource endpoint. Uses AWS PrivateLink.
 - AWS Marketplace services:** Connect to SaaS services that you have purchased through AWS Marketplace with an interface endpoint.
 - Service networks:** Connect to VPC Lattice service networks with a Service network endpoint. Uses AWS PrivateLink.
- Services (1/2):** A section with a search bar and a table of services.
 - Search:** A text input field with the value 'com.amazonaws.us-east-1.s3'.
 - Clear filters:** A button.
 - Table:** A table with columns: Service Name, Owner, Type, and Service Region.

Service Name	Owner	Type	Service Region
☒ com.amazonaws.us-east-1.s3	amazon	Gateway	us-east-1
☐ com.amazonaws.us-east-1.s3	amazon	Interface	us-east-1

Network settings

Select the VPC in which to create the endpoint.

VPC

Create the VPC endpoint in the VPC in the same AWS Region from which you will access a resource.

vpc-0e74d84df4d37625e (VPC-NV)

Route tables (1/3) [Info](#)

Search

< 1 >

☐	Name	Route Table ID	Main	Associated Id
☐	Private-RT	rtb-068f5be7dc53ace78 (Private-RT)	No	subnet-0bac41b4080c041e2 (Private-SN)
☐	-	rtb-079d55ea3c6b084b9	Yes	-
☒	Public-RT	rtb-0d36b3ec04fd6743a (Public-RT)	No	subnet-0201b003f65d6051c (Public-SN)

When you use an endpoint, the source IP addresses from your instances in your affected subnets for accessing the AWS service in the same region will be private IP addresses, not public IP addresses. Existing connections from your affected subnets to the AWS service that use public IP addresses may be dropped. Ensure that you don't have critical tasks running when you create or modify an endpoint.

[rtb-0d36b3ec04fd6743a](#) **Policy** [Info](#)

VPC endpoint policy controls access to the service.

☒ Full access

Allow access by any user or service within the VPC using credentials from any Amazon Web Services accounts to any resources in this Amazon Web Services service. All policies — IAM user policies, VPC endpoint policies, and Amazon Web Services service-specific policies (e.g. Amazon S3 bucket policies, any S3 ACL policies) — must grant the necessary permissions for access to succeed.

☐ CustomUse the [policy creation tool](#) to generate a policy, then paste the generated policy below.

```

root@ip-192-168-1-134:/home/ec2-user
[root@ip-192-168-1-134 ec2-user]# aws s3 ls
2025-10-12 18:34:58 s3-endpoint-bucket-nv-test
[root@ip-192-168-1-134 ec2-user]# |

```

Verification:

- From the Private-Server, run the following command:
aws s3 ls

Output:

- 2025-10-12 18:34:58 s3-endpoint-bucket-nv-test